

Similarity-Augmented Roadside Object Triangulation from street-level images

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Abstract

Object geolocation from street-level imagery under weak supervision is challenging due to noisy detections and the absence of bounding-box annotations. This work addresses this task via a similarity-augmented MRF framework that integrates visual consistency into geometry-driven multi-view triangulation. Starting from weakly supervised object detection, this formulation enforces both visual and geometric consistency across paired views to estimate object locations. To discourage visually inconsistent configurations, a Siamese network is trained to estimate pairwise visual similarity. Label noise introduced by weak supervision is addressed using a Beta-distribution-based filtering strategy during Siamese training. We also propose a novel deterministic optimizer for the MRF model that significantly reduces computational requirements and improves reproducibility. Experiments on real-world street-level datasets from Amsterdam and Dublin demonstrate consistent improvements in precision, recall, and localization error over a geometry-only baseline. The implementation will be made public upon acceptance.

Keywords: Siamese networks, Beta distribution, object geolocation, image similarity, street-level imagery

1 Introduction

Accurate geolocation of roadside objects from street-level imagery is essential for applications such as urban mapping, infrastructure monitoring, and autonomous navigation. Unlike image-level geolocation, object-level geolocation requires estimating precise real-world object coordinates from multiple views under significant uncertainty in depth, bearing, and appearance. Weakly supervised detectors [1] further amplify this challenge by producing uncertain object locations that generate excessive amounts of low-confidence triangulation candidates. Recent work has therefore shifted toward geometry-driven multi-view inference under weak supervision, where object locations are estimated from noisy geometric cues [2]. While video-based and track-driven approaches improve stability by exploiting temporal continuity [3], they inherently assume ordered observations and reliable tracking assumptions that do not hold in large-scale, crowdsourced image datasets.

Classical learning-based cross-view geolocation estimates camera locations by matching ground-level images to georeferenced aerial or satellite imagery using learned metric representations [4–6]. While effective, these methods rely on strong appearance cues and dense georeferenced databases, making them impractical for object-level localization of custom roadside assets. Beyond purely geometric formulations, recent works [7, 8] explore joint object association and localization in 3D using tracking and graph-based reasoning. Such approaches rely on dense image streams and strong calibration, limiting their applicability to sparser street-level imagery. Recent foundation models such as Segment Anything Model (SAM) and DINO-based methods can improve object localization and visual feature extraction, while DUST3R [9], and VGGT [10], can support depth estimation and multi-view reconstruction. However, their use for globally geolocating roadside objects from sparse views with uncertain cross-view associations remains underexplored.

More recently, GNNs have been adopted to explicitly model relational structure among views, objects, or observations, enabling learned aggregation of geometric and contextual cues [11]. Nassar *et al.* [12] formulate

localization as a graph inference problem, in which spatial relationships among observations are encoded as edges and refined via message passing to improve global consistency. While effective in structured settings, most graph-based approaches require dense supervision or explicit object grouping.

In contrast, purely geometric triangulation-based approaches are prone to failure under dense view coverage, where accumulated measurement uncertainty and ambiguous geometric configurations degrade localization accuracy. To address this, object geolocation is formulated as an energy minimization problem within an MRF, enforcing multi-view geometric consistency through unary and pairwise terms [13]. While effective, this formulation relies primarily on geometric cues, which limit its ability to suppress visually inconsistent yet geometrically plausible configurations in weakly supervised settings. Under weak supervision, MRF-based triangulation groups detections into view pairs that may include both true multi-view observations and spurious pairs arising from false positives [2]. Since both satisfy geometric constraints, geometry alone is insufficient.

In our work, we integrate visual consistency into the MRF framework. Specifically, we train a Siamese network to estimate pairwise visual similarity and propose a novel Beta-distribution-based filtering strategy to remove visually inconsistent pairs during training. The resulting similarity scores are incorporated as an additive energy term in the MRF, penalizing unreliable configurations. We introduce a novel deterministic optimizer for the MRF model that significantly reduces computational requirements and boosts reproducibility. Experiments with real-world street-level imagery demonstrate consistent improvements in object geolocation accuracy.

2 Methodology

This work proposes a noise-aware geolocation framework that improves the performance of multi-view triangulation under weak supervision. The proposed method retains weakly supervised object detection and an MRF-based triangulation framework, while explicitly addressing false positives introduced by weak supervision via similarity-driven noise filtering, as shown in Fig. 1. A learned image similarity term is incorporated into the MRF inference at the inference stage to enhance robustness and geolocation accuracy; see Fig. 2.

In the absence of bounding-box annotations, we adopt the weakly supervised detection and MRF-based geolocation pipeline presented in [2], corresponding to Parts A and B of Fig. 1. Although this framework enables scalable roadside object detection and geolocation, it inevitably produces false-positive detections. Some of these false positives may satisfy the geometric constraints of the MRF and form apparently valid multi-view pairs, leading to incorrect triangulations and reduced geolocation accuracy. To address this limitation, the multi-view image pairs generated by the MRF are processed using the similarity-learning framework shown in Fig. 1C. A Siamese network extracts feature representations from each image pair and computes their cosine similarity. Since the generated pairs may contain incorrect cross-view associations, the similarity scores are modelled using a two-component beta-mixture distribution to identify and remove noisy pairs. The resulting filtered dataset is then used to train the noise-aware similarity model. The trained weakly supervised detection and similarity models are then employed at inference (Fig. 2) for similarity-augmented roadside object geolocation.

2.1 Noise filtering

To address the presence of invalid view pairs, we employ a Siamese network to learn similarity-based relevance between image pairs. The network consists of two weight-sharing CNN branches for visual feature extraction, followed by a fully connected neural network (FCNN) that serves as the classification head. The candidate image pairs generated by the MRF-based geolocation framework are subsequently used to construct the training data for the similarity estimation task. The training data is divided into positive pairs (class 1), formed from images associated with the same object, and negative pairs (class 0), formed from images of different objects. Due to detection noise, the positive class inherently contains both clean pairs corresponding to true multi-view observations and noisy pairs arising from false detections.

Each image pair (v_i, v_j) is passed through two weight-sharing branches of the network, producing deep feature embeddings $f(v_i)$ and $f(v_j)$. The cosine similarity s_{ij} between the embeddings is computed as: $\text{sim}(x, y) = x \cdot y / (\|x\| \|y\|)$, where, $s_{ij} \in [0, 1]$. To distinguish between genuine and noisy image pairs within

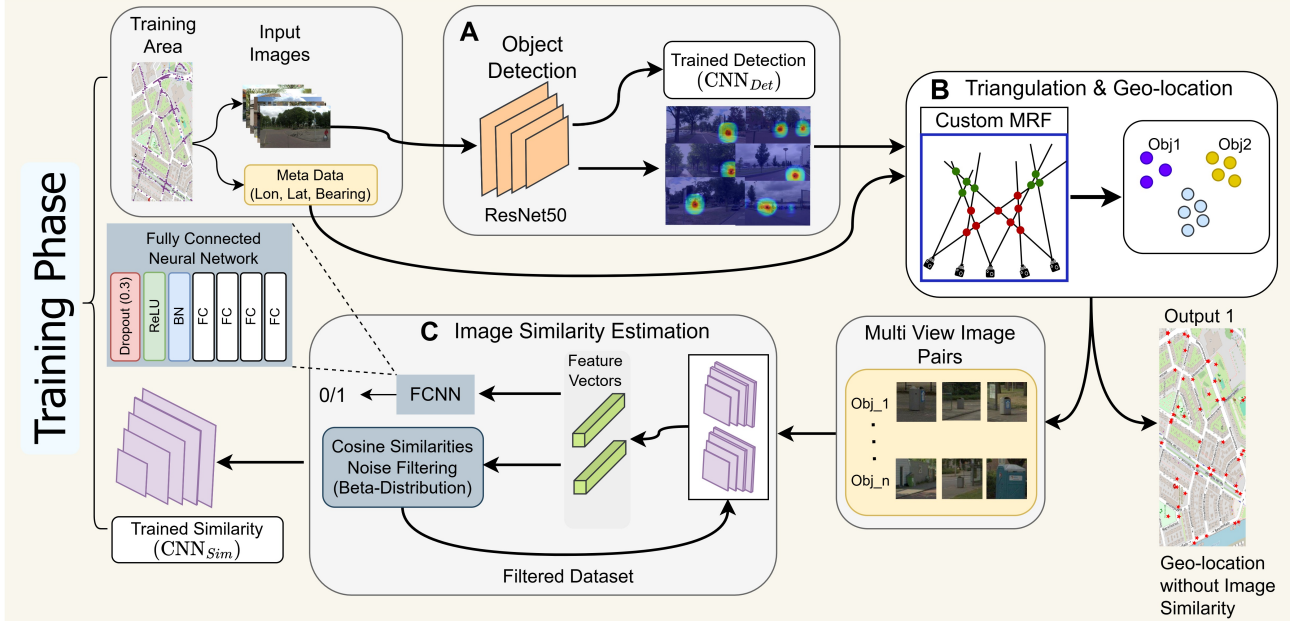


Figure 1: Proposed training pipeline: (A) weakly supervised object classification and Grad-CAM-based localization, (B) MRF-based triangulation to generate multi-view image pairs, and (C) Siamese similarity learning with Beta-mixture filtering.

class 1, we model the distribution of similarity scores using a two-component Beta mixture model, defined as

$$p(s) = \omega \text{Beta}(s | a_1, b_1) + (1 - \omega) \text{Beta}(s | a_0, b_0), \quad (1)$$

where $\text{Beta}(\cdot)$ denotes the Beta probability density function. The component $\omega \text{Beta}(s | a_1, b_1)$ captures high-similarity, visually consistent image pairs, while $(1 - \omega) \text{Beta}(s | a_0, b_0)$ models low-similarity pairs corresponding to noisy or mismatched observations. We adopt the Beta distribution for this task because the similarity scores are naturally constrained to the interval $[0, 1]$. Moreover, its flexibility allows it to model a wide range of distributional shapes, including uniform, skewed, and unimodal forms, making it an appropriate choice for characterizing the unknown similarity distributions associated with both genuine and noisy pairs.

The mixture parameters (a_k, b_k, ω) are estimated using the Expectation-Maximization (EM) algorithm, which is well-suited for mixture models with latent component assignments. EM alternates between estimating posterior probabilities and updating the mixture weights and Beta parameters. This iterative nature enables stable convergence despite overlapping components. More complex alternatives, such as stochastic or simulation-based Bayesian methods, are unnecessary in our setting due to the reliable convergence of the standard EM formulation. The EM algorithm follows the following two alternating steps to estimate and update the mixture parameters:

E-step: Given the current parameter estimates, we compute the posterior probabilities for any s_i as:

$$\gamma_0(s_i) = \frac{\omega_0 \text{Beta}(s_i; a_0, b_0)}{\omega_0 \text{Beta}(s_i; a_0, b_0) + \omega_1 \text{Beta}(s_i; a_1, b_1)}, \quad (2)$$

where ω_0 and $\omega_1 = 1 - \omega_0$ denote the mixture weights. The posterior probability of the second mixture component is obtained as the complement of the first, i.e., $\gamma_1(s_i) = 1 - \gamma_0(s_i)$.

M-step: Using the posteriors from the E-step, samples are assigned to the component with the higher posterior probability, i.e., $\arg\max_k \gamma_k(s_i)$. The shape parameters of each Beta component are then updated via maximum likelihood estimation as $(a_k, b_k) = \text{MLE}_{\text{Beta}}(\{s_i | \gamma_k(s_i) \geq 0.5\})$, $k \in \{0, 1\}$, where $\text{MLE}_{\text{Beta}}(\cdot)$ denotes Beta parameter estimation from the assigned samples.

Starting from a trivial initialization, the parameters α_k and β_k are refined iteratively. The mixture weight ω_k , which represents the current estimate of the proportion of samples associated with component k , is updated at

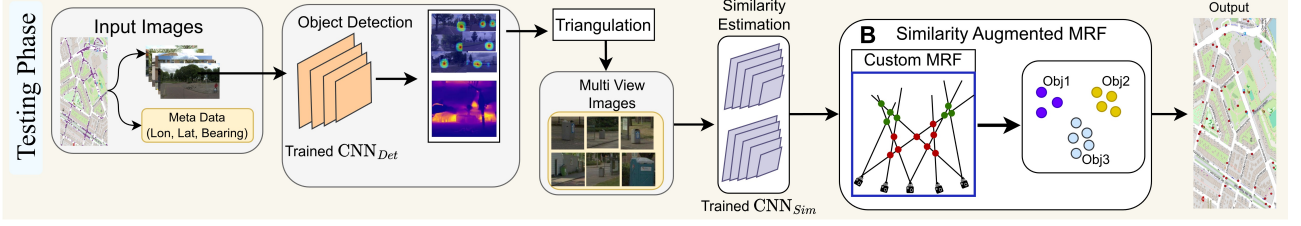


Figure 2: Inference pipeline: object detection and localization, triangulation-based view pairing, Siamese similarity estimation, and final geolocation using a similarity-augmented MRF.

each iteration, based on the posterior probability. Specifically, it is computed as:

$$\omega_k = \frac{|\{s_i \mid \gamma_k(s_i) \geq 0.5\}|}{|D|}, \quad k \in \{0, 1\} \quad (3)$$

where $|D|$ denotes the total number of samples, this update reflects the fraction of the dataset currently attributed to each mixture component.

The proposed training strategy alternates between similarity learning and noise filtering. We first train the Siamese network for K epochs to learn an initial similarity representation from the full dataset. Subsequently, the noise filtering procedure is applied to identify and remove unreliable pairs, and the network is fine-tuned on the filtered dataset for an additional K epochs. This staged fine-tuning, both before and after noise filtering, helps ensure that hard positive examples are learned by the model rather than incorrectly removed as noisy pairs. This alternating process of filtering and fine-tuning is repeated iteratively for C cycles. Training is terminated once the estimated proportion of filtered samples, ω_k , falls below a predefined threshold of 2%, and the final fine-tuned model is retained for evaluation.

2.2 Loss functions

The detection network is trained as a binary classifier using the BCE loss $\mathcal{L}_{CE}$. For similarity learning, a Siamese network is optimized with a joint loss comprising classification and metric objectives. The FCNN classification head is supervised using the cross-entropy loss $\mathcal{L}_{CE}$, while the shared CNN embeddings are regularized using a contrastive loss ($\mathcal{L}_c$),

$$\mathcal{L}_c = \frac{1}{N} \sum_{i=1}^N [y_i d_i^2 + (1 - y_i) \max(0, m - d_i)^2], \quad (4)$$

and cosine embedding loss ($\mathcal{L}_{COS}$),

$$\mathcal{L}_{COS} = \frac{1}{N} \sum_{i=1}^N \begin{cases} 1 - \cos(\mathbf{x}_i, \mathbf{y}_i), & y_i = 1, \\ \max(0, \cos(\mathbf{x}_i, \mathbf{y}_i) - m), & y_i = 0, \end{cases} \quad (5)$$

where d_i denotes the Euclidean distance between embeddings $(\mathbf{x}_i, \mathbf{y}_i)$ and m is the margin parameter.

The overall Siamese training objective is given as a weighted sum of the three contributing losses with a single weight: $\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{COS} + (1 - \lambda) \mathcal{L}_c$.

2.3 Information fusion with MRFs

Object detections in images combined with the image metadata are used to construct rays (in 2D) that originate at the camera's GPS coordinates (assumed identical to the camera center). The camera's viewing direction is obtained from the heading provided in the image metadata, while the object's position in the image is used to calculate its final bearing. The bearing (direction of the ray) is selected towards the middle of the segmented object. The rays create a 2D graph with nodes formed by the intersections of these rays. Edges are established between nodes on the intersected rays. An MRF model is employed to estimate binary decisions ($z_i \in \{0, 1\}$)

indicating the presence of the object of interest at a particular node i [13]. The MRF energy of a node z_i is governed by two geometric unary terms and one pairwise term. We extend this formulation by introducing an additive unary term u_2 , which incorporates visual information by rewarding or penalizing a node based on the Siamese similarity of the two views that generate it. Thus, the total energy is defined over all $N_{\mathcal{Z}}$ nodes as:

$$\mathcal{U}(\mathbf{z}) = \sum_{i=1}^{N_{\mathcal{Z}}} \sum_{j=0}^2 \alpha_j u_j(z_i) + (1 - \sum_{j=0}^2 \alpha_j) \sum_{i=1}^{N_{\mathcal{Z}}} \sum_{j=1}^{N_{\mathcal{Z}}} p(z_i, z_j) \quad (6)$$

The first unary term penalizes the discrepancy between the distance from the camera to the node estimated with monocular depth CNN (Δ_{ij}) and with triangulation (d_{ij}): $u_1(z_i) = z_i \sum_{j=1,2} \|\Delta_{ij} - d_{ij}\|$. The u_0 unary term promotes inclusion of terms on any ray R and is inversely proportional to the number of intersections on the ray: $u_0(z_i) = -z_i / (\sum_{x_n \in R} 1)$. The presence of this term ensures that the minimum energy is not reached on the empty configuration $z_i = 0$ for all i . The pairwise term penalizes distances between any two nodes simultaneously active on any view-ray R : $p(z_m, z_n) = \sum_{x_m, x_n \in R} z_m z_n \|x_m - x_n\|$, where x_i is a GPS position of node controlled by state z_i . The similarity unary term u_2 penalizes nodes generated from visually inconsistent view pairs: $u_2(z_i) = z_i(1 - s_i)$, where $s_i \in [0, 1]$ denotes the similarity score predicted by the Siamese network. Nodes supported by visually consistent views ($s_i \approx 1$) incur minimal penalty, while intersections formed by visually dissimilar views are penalized, thereby suppressing spurious nodes that are geometrically admissible but visually inconsistent. The contributions of the terms are governed by $\alpha_i \in [0, 1]$, $\sum \alpha_i < 1$.

The state $\mathbf{z} = \{z_i\}_{i=1, \dots, N_{\mathcal{Z}}}$ that minimizes the energy $\mathcal{U}$ (Eq. 6) is computed using the Quadratic pseudo-Boolean optimization (QPBO) algorithm [14], which is based on graph-cut optimizations. QPBO has two substantial advantages over the original Simulated Annealing-based optimization proposed in [13]: firstly, the optimization result is deterministic, which avoids the need to rerun the procedure multiple times to arrive at an averaged (stable) field configuration. Secondly, the procedure has predictable complexity and is fast. The use of QPBO is possible by eliminating the higher-order penalty terms originally used in [13], which did not allow energy function composition compatible with this type of optimization. Specifically, the stand-alone penalty term u_3 used in [13] has been replaced by the unary term u_0 (Eq. 6). The pairwise term is supermodular [14], but the number of such terms is relatively small, and empirically allows for optimal state assignments to all graph nodes using the QPBO/QPBOP algorithm.

3 Datasets and Experimentation

3.1 Dataset Description

Although several multi-view object geolocation datasets have been proposed, no widely accepted benchmark or evaluation protocol exists [15]. Existing results remain dataset- and implementation-specific, while relevant datasets such as Pasadena Multi-View [12] are not publicly accessible. Here, we employ two publicly available street-level datasets from Dublin and Amsterdam, using public waste bins as the target objects, to support future comparisons.

Dublin. street-level images are obtained via the Mapillary API ¹, selecting only images flagged as containing wastebins. Public street furniture data from the Smart Dublin bins dataset ² is used for spatial supervision: images within a 15 m radius of DCC bin locations are labeled as class 1 (9,900 images), while class 0 samples are selected from images located 30-100 m away from any bin. All images are resized to 1500 × 900. We evaluate on two test sets from Mapillary and the DCC dataset, covering 2.7 km of roads with images sampled every 3 m, six 68.77° views per location.

Amsterdam. Panoramic images are obtained via the Mapillary API, restricted to city-provided panoramas flagged as containing wastebins and captured with a fixed north-facing orientation (0°). The ground-truth locations of wastebins are obtained from the OSM API ³ and used to associate panoramas with object positions.

¹Mapillary: <https://www.mapillary.com>

²Smart Dublin: <https://data.smartdublin.ie/dataset/dcc-public-bin-locations>

³OpenStreetMaps: <https://www.openstreetmap.org>

Table 1: Geolocation performance on the Amsterdam and Dublin datasets against MRF [13]

Dataset	Method	Precision (%)	Recall (%)	Error (m)
Amsterdam	MRF	64.25	79.9	2.54
	Similarity-Aug. MRF	68.2	86.1	1.87
Dublin (Mapillary)	MRF	66.21	80.3	2.51
	Similarity-Aug. MRF	71.4	87.1	1.52
Dublin (DCC)	MRF	65.1	80.8	2.38
	Similarity-Aug. MRF	72.8	87.9	1.49

Table 2: Classification results on the Amsterdam dataset

Model	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)
ResNet-34	94.7	88.2	91.3	92
ResNet-50	95.81	89.10	92.33	93.1
Resnet-101	93.1	88	90.47	92.4

A total of 4,100 panoramas (8000×4000) are split into 2000×2000 sub-images and labeled using Mapillary masks, yielding 20,000 samples, of which 85% are used for training and 15% for testing.

3.2 Training Setup

The training follows a two-stage setup comprising weakly supervised object detection and Siamese-based similarity learning, with MRF-based geolocation performed at inference time.

Backbone architectures. ResNet-50 is used as the primary image classification backbone, with Grad-CAM employed to localize the object within each image. MobileNetV2 is adopted as the backbone feature extractor in the Siamese architecture due to its proven effectiveness in metric learning-based similarity estimation [16].

Implementation details. All models are implemented in PyTorch and trained on an NVIDIA Titan X GPU (12 GB). The detection network is trained for 20 epochs using Adam with a batch size of 8 and a step learning-rate scheduler (step size 5, $\gamma = 0.1$). For similarity learning, the Siamese network employs a MobileNetV2 backbone with an FCNN head (512, 512, 256, 128), batch normalization and ReLU activations, and dropout (0.3) applied to the final two layers. Training uses the Adam optimizer with an initial learning rate of 10^{-3} , a batch size of 32, and a learning-rate decay every 5 epochs. Optimization alternates between Siamese training and Beta-based pair filtering, with $K \in [3, 7]$ epochs per cycle until validation accuracy stabilizes, and terminates when the estimated noisy-pair ratio ω_k falls below 2%. The combined loss is optimized with $\lambda = 0.45$, identified via a grid search.

3.3 Results Analysis

We report results for weakly supervised object detection and geolocation on the Dublin and Amsterdam datasets, including impact analysis of Beta-distribution-based noise filtering.

Object detection performance. ResNet-34, ResNet-50, and ResNet-101 are evaluated for image classification on the Amsterdam dataset. As shown in Table 2, ResNet-50 achieves the best overall performance, with 95.81% precision, 89.10% recall, and 93.1% accuracy, and is therefore selected as the primary classification backbone. Using the selected ResNet-50 backbone, classification performance across all test sets is reported in Table 3. On the Amsterdam dataset, the model achieves 95.81% precision, 89.10% recall, and 93.1% accuracy. Comparable performance is observed on the Dublin datasets, with 93.1% precision, 87.2% recall, and 92.8% accuracy on Dublin (DCC), and 94.21% precision, 89.0% recall, and 93.0% accuracy on Dublin (Mapillary). This indicates that the detection model generalizes well across cities and data sources, providing reliable geolocation.

Effect of noise filtering. We next analyze the effect of the proposed Beta-distribution-based noise filtering on image pairs generated after detection and MRF-based triangulation. Approximately **8.5%** of pairs are filtered in the Dublin dataset and **12.3%** in the Amsterdam dataset, reflecting greater visual ambiguity in panoramic

Table 3: Classification results with Resnet-50 architecture

Test Set	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)
Amsterdam	95.81	89.10	92.33	93.1
Dublin (DCC)	93.1	87.2	90.05	92.8
Dublin (Mapillary)	94.21	89	91.53	93



Figure 3: Filtered image pairs after Beta-distribution-based denoising in Dublin (top) and Amsterdam (bottom)

imagery (Fig. 3). In the absence of ground-truth pair labels, manual inspection indicates that about **85%** of the removed pairs correspond to genuine mismatches. During inference, the pretrained Siamese network consistently assigns low similarity scores to the filtered pairs, confirming the effectiveness of denoising.

Geolocation performance. We now evaluate the geolocation performance of the proposed framework on the Amsterdam and Dublin datasets. At inference, images are processed by the pretrained detection model, triangulated to generate admissible intersections, and refined using similarity scores from the Siamese network.

Table 1 reports the geolocation results in terms of precision, recall, and absolute localization error. Compared to the baseline MRF, incorporating similarity cues consistently improves performance: on the Amsterdam dataset, precision increases by 4% and recall by 6.2%, while the mean localization error is reduced by 0.7 m; similar trends are observed on the Dublin datasets. These results show that integrating similarity-aware filtering into the MRF effectively suppresses spurious intersections arising from noisy detections, leading to higher precision and recall and substantially lower localization error. Overall, the proposed similarity-augmented MRF yields more robust and accurate geolocation across three diverse datasets. The QPBO solver for MRF reduces computational complexity by an order of magnitude and yields deterministic, and therefore reproducible, geolocation results.

4 Conclusion

This work presents a similarity-augmented MRF framework for object geolocation under weak supervision. By integrating Siamese-based visual similarity into a geometry-driven triangulation model and mitigating label noise via Beta-distribution-based filtering, the proposed method effectively suppresses false-positive detections. We deploy a new deterministic optimizer for MRF, resulting in more stable and computationally feasible results. Experiments on real-world street-level datasets from Amsterdam and Dublin show consistent gains in precision, recall, and localization accuracy over a geometry-only MRF baseline. Future work will focus on joint optimization of detection and similarity learning.

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